

SUMMARY STATEMENT**PROGRAM CONTACT:****(Privileged Communication)****Release Date: 03/30/2022****Revised Date:**

Application Number: 1 I01 HX003625-01**Principal Investigator****ZEVALLOS, JOSE P****Applicant Organization: ST. LOUIS VA MEDICAL CENTER****Review Group: HSR3
HSR-3 Healthcare Informatics****Meeting Date: 03/02/2022
Council: MAY 2022
Requested Start: 07/01/2022****RFA/PA: HX22-001**

**Project Title: Understanding Barriers to HPV Vaccination and their Impact on HPV+
Oropharyngeal Cancer Incidence and Outcomes in the US Veteran Population**
SRG Action: Impact Score:287 Percentile:73.4 +**Human Subjects: 30-Human subjects involved - Certified, no SRG concerns**
Animal Subjects: 10-No live vertebrate animals involved for competing appl.
Gender: 1A-Both genders, scientifically acceptable
Minority: 1A-Minorities and non-minorities, scientifically acceptable
Age: 1A-Children, Adults, Older Adults, scientifically acceptable
Clinical Research - not NIH-defined Phase III Trial

Project Year	Direct Costs Requested
1	284,328
2	289,088
3	292,801
4	296,588
TOTAL	1,162,805

ADMINISTRATIVE BUDGET NOTE: The budget shown is the requested budget and has not been adjusted to reflect any recommendations made by reviewers. If an award is planned, the costs will be calculated by VA Office of Research and Development (ORD) staff based on the recommendations outlined in the BUDGET COMMENT section and any relevant ORD service-specific limitations.

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Key Summary Points:

1. Impactful and innovative area of study.
2. Strong experienced team.
3. Details for how applicant will address the competing guidelines is lacking (ACS and USPSTF)
4. Disconnect between Aims. How will Aims 1 and 2 inform the Markov Model and potential gaps of the model.
5. Details about data sources and feasibility for constructing the measures are lacking. Pilot data would be useful to demonstrate ability to work with the data proposed in the study.
6. Veteran engagement could be better incorporated within the proposal and approach.

DESCRIPTION (provided by applicant):

Background: Oropharyngeal cancer (OPC) caused by oncogenic human papillomavirus (HPV) is increasing rapidly in incidence in the general population of the United States and at an even more rapid pace among US Veterans.

Significance: There are currently no extant estimates of HPV (+) OPC burden in the Veterans Health Administration (VHA) which makes it exceedingly difficult to develop prevention strategies aimed at Veterans. Although HPV vaccination holds the promise of primary prevention of HPV-associated cancers, recent data suggest that vaccination rates in active-duty military and Veteran populations are extremely low. The Center for Disease Control recently expanded HPV vaccination indications to include men and women <45 years of age, making many Veterans entering the VA healthcare system eligible. Veterans smoke at a higher rate compared to the general population, and it is also known that tobacco use diminishes prognosis in HPV (+) OPC.

Objectives/Hypothesis: The objective of this proposal is to understand the current and future burden of HPV (+) OPC in the VHA population by determining the prevalence of HPV vaccination among eligible Veterans, measuring disease incidence and outcomes in a national VHA HPV (+) OPC cohort, and developing a tool to model the potential benefit of primary and secondary prevention on HPV (+) OPC incidence and mortality. Innovation and Impact: This is the first study to address barriers to HPV vaccination in the VHA community. Completion of this study will generate a clinical meaningful impact on individual Veterans at risk for HPV (+) OPC and inform the development and implementation of impactful prevention strategies. Specific Aims: 1. To define the prevalence of HPV vaccination among eligible Veterans and identify populations at highest risk for under-vaccination, 2. To measure the impact of tobacco exposure on HPV (+) OPC outcomes in a contemporary VHA cohort, 3. To project future HPV (+) OPC incidence and cancer-related mortality in the VHA.

Methodology: Administrative database analysis using VHA Corporate Data Warehouse (CDW) to determine prevalence of HPV vaccinations, retrospective cohort of Veterans with HPV (+) OPC, Markov simulation modeling to project future incidence, outcomes, and preventive interventions for HPV (+) OPC.

Next Steps/Implementation: The results of this study could be used to inform VISN-specific primary (HPV vaccination) and secondary (smoking cessation) in order to decrease the future burden of HPV (+) OPC among Veterans.

CRITIQUE 1**1. Significance.**

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This proposal addresses a cancer control issue important to VA health care. HPV+OPC leads to significant morbidity and mortality and Veterans are at high risk due to higher risks of smoking and lower risks of HPV vaccination than the general population. Further, the CDC has recently increased the upper age of vaccination to 45 years, offering the opportunity for Veterans receiving care in the VA to obtain the vaccine if not received previously. Smoking cessation is also important for this population both with respect to morbidity and mortality associated with smoking among HPV-OPC patients. The proposed work Aims to describe the burden of this condition among Veterans and the potential to decrease morbidity and mortality associated with this condition through vaccination uptake and smoking cessation efforts. In the area of cancer control this work could raise the visibility of this issue and inform policies and interventions with the potential for high impact.

The overall goals of this project are 1) to define the prevalence of HPV vaccination among eligible Veterans and identify populations at highest risk for under-vaccination, 1) to measure the impact of tobacco cessation on HPV+OPC outcomes in a VHA cohort, and 3) to project future HPV +OPC incidence and cancer related mortality in the VHA.

2. Innovation and Impact.

The study is innovative in focusing on factors associated with low uptake of HPV vaccination in the VHA community. Its methods are innovative in combining designs of a retrospective cohort study using VHA data, a retrospective cohort study matching Veterans to a non-Veteran cancer cohort, and Markov modeling to assess the decreased burden of disease due to HPV+OPC anticipated if prevention (vaccination and smoking cessation) can be addressed. The premise of the study is also based on a concern that it could be unsafe to have apply de-escalation of treatment for HPV+OPC among Veterans as compared to the general population because of data indicating more aggressive disease among Veterans, particularly associated with Veterans who are heavy smokers.

The study strength is in describing and projecting the burden of disease and highlighting the impact of increased adoption of prevention strategies. However, this study does not directly inform the interventions that could be undertaken to increase adoption of prevention methods. It is anticipated that the findings will increase the prioritization of this problem among VHA leaders. Further work, as alluded to in the proposal, including Veteran and provider engagement will be needed to address barriers to uptake of proven prevention methods.

The implementation plan outlines short term and long term implementation strategies. Strategies focus on the support of future interventions and clinical trials related to vaccination and smoking cessation to decrease the burden of HPV+ OPC in Veteran populations. A focus is on health policy at VISN and VHA level in response to findings from this work.

3. Approach.

Aim 1 Critique

1. Lack of pilot data presented on ability to collect vaccination status.
2. Lack of data on the sensitivity and specificity of cited algorithms used to collect smoking status with the VA CDW database. The investigators recognize the limitations of obtaining accurate smoking history and suggest an alternative measure based on interpretation of CT scans among those that have had a chest CT scan as part of a staging evaluation. However, all patients in Aim 1 may not have had a chest CT scan.

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3. Provide additional justification and consideration of SDOH or other factors related to hypothesized lower vaccination rates by race/ethnicity. Currently, the lower rates among African Americans cited from civilian populations is only reason provided for this hypothesis. Given expertise in disparities on the research team, a more in depth evaluation of these associations would be valuable as it relates to the statistical analysis plan and interpretation of data.
4. Unclear if incidence of HPV related cancers will also be analyzed in Aim1. Although the rates will be low in the cohort with age < 45, some Veterans may have presented with HNCA or other HPV related cancers in this age group.

Aim 2 Critique

1. The hypothesis is based on >30 pack years exposure will have a higher impact on PFS and OS among Veterans than the general population. Threshold of 30, 30-60, and > 60 pack years may not capture important differences by smoking exposure. Is there pilot data to support this distribution of smoking history at 40%, 40%, and 20% in the Veteran population?
2. Specify what files within VA CDW files are being used to determine cancer clinical indicators including curative intent treatment and pathology diagnosis and other inclusion/exclusion criteria for this cohort. Do the investigators have pilot data from these sources to confirm their ability to obtain this data?
3. Similarly, what data will be used to determine loco regional control, distant metastasis, and progression free survival among Veterans?
4. Additional detail regarding the matching procedures would strengthen this Aim. What was the justification for using the proposed matching variables and what is the ability of investigators to extract those variables for both cohorts?

Aim 3 Critique

1. Provide details on how smoking status is incorporated into transition probabilities, i.e., is it in terms of pack-years, years of smoking, or a dichotomous variable of current smoking status?
2. Clarify if and how data from Aim 1 and Aim 2 be incorporated into the transition probabilities for the Markov model.

Critique of Veteran Engagement

1. Veteran engagement section: Results leading to qualitative research in this area. Although stated they will work with VISNs to develop Veteran focus groups and interviews to understand barriers-this is not included in the proposed work. Also stated that after Aim 2 they will seek input from Veterans regarding attitudes on treatment de-intensification that may inform clinical trials. Similarly, stated that Veteran perceptions will be sought based on results of Aim 3. However, there is no qualitative work or Veteran engagement described in the methods of this

4. Feasibility.

Adding pilot data that used the databases cited would strengthen support for feasibility of the project.

5. Implementation.

Results may inform VISN specific primary prevention (HPV Vaccination) and secondary (smoking cessation) to decrease burden of HPV+OPC among Veterans. The study also supports national VA priorities in improving access to care, addressing health disparities, and promoting population health. Although methods and analyses incorporate race, there are no analyses articulated to further understanding the mechanism of race related disparities.

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It's unclear how a risk stratification nomogram will address potential role for treatment de-intensification in VA-this idea needs further development.

Markov model-may help to describe and highlight the anticipated burden of OPC among Veterans in coming decades. However, the model requires greater specification of the transition states.

6. Investigator Qualifications.

The team is excellent and has the experience to complete this work.

Dr. Zevallos, MD, MPH Division Chief of Head and Neck Surgical Oncology and Associate Professor at Washington University SOM. Has pilot VA funding for epidemiology of OPC among Veterans. Has current R01 NCI funded project in HNCA.

Dr. Sandulache, MD, PHD-ENT physician-will provide clinical expertise for study, investigator in smoking related SCC. Will work on Veteran engagement strategies.

Drs. Zevallos and Sandulache have an established history of working together.

Dr. Mazul (5%)-HNCA epidemiologist cancer disparities and Assistant Professor ENT at WUSM. NIH Funded in HNCA disparities study.

Dr. Chang, PHD, SM (15%effort)-Associate Professor in Division of Public Health Services, Department of Surgery, WUSM. Applied econometrician and HSR researcher. Experienced researcher with prior funding by AHRQ, NIH. Extensive experience with research in VHA databases. Experienced HSR researcher in methods using VHA administrative data and with simulation models.

Also with adequate support listed in research coordinator, data analyst, decision modeler.

7. Multiple PI Leadership Plan.

Not applicable.

8. Facilities and Resources.

The resources page highlights Washington University School of Medicine, the Department of Surgery at WUSM, and The Division of Public Health Sciences within the Department of Surgery all of which are very strong and supportive of the proposed work. Also highlights the VA St Louis Health Care System and establishing the computing resources needed for the work proposed. Would like to see integration with the Cancer Center at the WUSM.

9. Adequacy of Response to Previous Feedback Regarding the Proposed Study.

Not Applicable-a first submission.

10. Protection of Human Subjects from Research Risk.

Adequate.

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11. Inclusion of Women and Minorities in Research.

The study will include women and minorities.

12. Budget and Period of Support.

No issues noted with budget.

13. Sharing Research Data (Data Management and Access Plan (DMAP)).

Appropriate data sharing.

14. Overall Impression.

This is an innovative study with the potential for high impact in VA cancer control pertaining to HPV+OPC. The strengths include the use of a range of study designs to better understand factors associated with uptake of HPV vaccine among Veterans, factors associated with poor treatment outcomes of HPV+OPC among Veterans, and predictions for how uptake of HPV vaccines and smoking cessation among Veterans with HPV+OPC cancer can decrease the burden of disease among VHA. However, there are limitations including lack of integration of the specific Aims. The key strengths and limitations noted below.

15. Key Strengths.

1. Important topic for Veterans with potential of findings to inform future interventions in cancer control for HPV+OPC in VA populations. In particular, findings could advance the use of evidence-based prevention among Veterans (HPV vaccine) and support the need for clinical trials among Veterans in de-escalation of treatment for HPV+OPC.
2. Excellent study team with needed skills and experience to conduct the work.
3. Aims address a gap in knowledge regarding prevention and clinical care for HPV+OPC among Veterans
4. A generally well written proposal

16. Key Weaknesses.

1. Major: Aim 1: Determining smoking history: Detailed smoking status limited in Health Factors from CDW-classified as former, current, and never on index date of first primary care visit in study period or within one year before index date. Unclear how investigators will obtain the detailed smoking history required to classify smoking history in following categories: 0-30 pack years, 30-60 pack yrs., >60 pack years. Alternative approach of estimating smoking history from CT scans may not work for Aim 1 as unclear how many subjects would have a CT scan. Pilot data would help to address this weakness.
2. Major: Aim 2: Specify where in CDW data investigators will obtain variables needed for match, such as T-classification, N-classification, and sub-site. Provide additional detail about the matching process and methods to be used.

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3. Minor Aim 2: Clarify that the 300 HPV+ patients provide adequate power for the multivariate analysis to evaluate impact of HPV/P16 status on survival and determine HR for tobacco exposure and HPV/p16 status.
4. Minor Aim 3: For simulations of smoking cessation, clarify if smoking is considered binary or if the level of smoking is considered in the model.
5. Major: Future directions and Impact. Unclear how findings from this work will direct targeted interventions. The development of these interventions for vaccine uptake and smoking cessation will require qualitative work and development of behavioral change or system interventions. Although this was mentioned in this proposal, there were no methods presented to conduct this work or to engage Veterans to determine strategies for conducting this work. The proposal would benefit from involvement of a Veteran Advisory Board or other mechanism for Veteran and stakeholder input.
6. Major-lacks integration across specific Aims. How are findings from Aim 1 and Aim 2 to inform the Markov model in Aim 3?

CRITIQUE 2

4. Significance.

Strengths

1. Proposal makes clear the overlapping risks for HPV and smoking, and that the rates of both are higher in the Veteran population, along with a lack of mandatory military vaccination for HPV, provide a situation in which understanding the risks for relevant patients is important.
2. Aim 2, any approach comments below aside, has the potential to generate meaningful and clinically actionable results.

Weaknesses

1. The application tries to deal with:
 - a. the unvaccinated opportunity population, in which almost all Vets presenting will be in the 26-45 age group, and in which the American Cancer Society expressly disagrees with the ACIP informed choice recommendations because of: 1) low effectiveness and low cancer prevention potential in this age group, 2) burden of decision making and uncertainty in patients and clinicians, and 3) lack of sufficient guidance on which individuals might be the most to benefit, such as those with no or only a few prior sexual partners.
 - b. And then moves the meter forward past years of HPV+ incubation to a OPC cancer cohort, which seems highly relevant to the VA because of low vaccination rates, higher risk seeking behavior, and higher smoking rates, along with a strong male rise in these risks.

5. Innovation and Impact.

Strengths

1. The proposal convinces that HPV+ OPC is a significant issue for Veterans in the VA, and with the uncertainty about vaccination of the 26-45 group, and the challenges around PFS and OS, there are potentially prevention and post-cancer management changes that might occur from a robust analysis along the state changes.

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2. There is an opportunity for some of this work to generalize to outside the VA
3. This could impact clinical practice if the results are robust enough.

6. Approach.

Strengths

1. Aim 2's letters of support and overall conceptual design of VA & non-va data seem appropriate and feasible to achieve the goal for that Aim after some methods details are clarified.

Weaknesses

1. Major: Aim 1 is a reasonable Aim 1 for a grant focused around HPV vaccination, extending the recent article on VA HPV vaccination to include 26-45 and including regional and racial sub-analyses, but does not conceptually fit with Aim 2 and 3. The subsequent Aims for that type of grant would need to try to figure out which populations might most benefit from the optional 26-45 vaccination, even before any work to look at barriers to actually getting vaccines to that group would be justified (see significance and ACS position). Aims 2 & 3 fit together and fit with the PI's general focus.
2. Major: Aim 1's focus on multivariate analysis to detect barriers to vaccination is premature in the context of the equipoise and clinical disagreement around vaccination 26-45, primarily around not being sure who is most likely to benefit... this is a question to answer for Vets before barriers. I also agree with reference 25's recommendation that the military should make it mandatory vaccination and/or this would be a much cleaner justification for a DoD research project (<26).
3. Major: Aim 2, Please be clear if this proposal is executing an association study looking for key risk factors or a risk prediction model for PFS and OS. There is language suggesting both. If association, no need for split training and testing data sets, and focus on risk factors. If prediction model, 10-fold cv or bootstrapping better than 2/3, 1/3, and also need to assess discrimination and calibration and potentially decay over time. If prediction, need to include considerations of building on matched cohort instead of a full cohort. Comments around Exp 2.2 nomogram also suggest risk prediction model.
4. Major: Aim 3 conceptual gap: it is not clear from this proposal how the data and work from prior Aims would facilitate modeling estimations of state transition from well to oral HPV because the HPV exposure status of unvaccinated Veterans is not known, and no discussion of the cumulative risk of HPV+ and/or smoking status exposures over time transition to HPV+ OPC, which are key conceptual areas that are the gap between Aim 1 and Aim 2. Stated another way, the proposal states transition probabilities will come from references 1, 4, and 47, but are not explicitly framed or justified, or described in a way that makes it clear how it will all fit together.
5. Minor: medSAS files are not the best accessed, better to use CDW sql files or OMOP data.
6. Minor: Aim 2, VA & NCDB matching, concern systematic cohort differences in Veterans would lead to unmeasured confounding with only age, TNM, sub-site matching, rather than something more potentially robust like propensity score matching with more covariates, etc. Please explicitly justify. Related is it is not clear which variables would be used in multivariable analysis; if the variables 'individually tested' in exp 2.1 are also the ones in multivariate, seems very sparse.
7. Minor: Aim 3: drawing from veteran population 2008-2009 likely to reflect very different population for the generated simulated cohort than more recent population. Presuming it's because you need time-surveillance for the model, but not explicitly justified.

4. Feasibility.

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The overall timeline and sample size estimates seem reasonable.
Staffing challenges with so many TBD positions.

5. Implementation.

Plans to share results with VISN and VHA OPC working group. Overall, operational partnership does not to be strongly integrated with proposal (letters of support).

6. Investigator Qualifications.

Strengths

1. Highly qualified team with expertise in the clinical conditions at hand, strong publication record, expertise in simulation modeling.

Weaknesses

1. Concern over so many TBD positions, given the challenges in hiring data scientists right now even outside the VA.

7. Multiple PI Leadership Plan.

Not applicable.

8. Facilities and Resources.

Adequate to execute the study as planned, no concerns.

9. Adequacy of Response to Previous Feedback Regarding the Proposed Study.

Not applicable.

10. Protection of Human Subjects from Research Risk.

Observational routine care data only, minimal risk to subjects, protections seem appropriate.

11. Inclusion of Women and Minorities in Research.

The grant does include data and consideration for both men and women, and seeks to evaluate potential disparities in care among racial sub-populations.

12. Budget and Period of Support.

Budget seems appropriate for the scope of work, period of support seems appropriate.

13. Sharing Research Data (Data Management and Access Plan (DMAP)).

#8: states that data sharing will allow validation studies in and out of the VA, not sure if VA data sharing outside the VA is implied by that statement.

14. Overall Impression.

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Overall, this is an interesting proposal that successfully sets the stage and need for additional research around Veterans with and at risk for HPV+ OPC. Aim 2 is particularly strong, and the critique in the methods in that section easily addressable. As put together, Aim 1 does not fit together well with Aims 2 and 3, and the stage from healthy -> HPV+ w/w/o smoking -> HPV+ OPC and working through explicitly all those stages is not clear or well described, which limits enthusiasm regarding Aim 3. Concerns about the disagreements in vaccination recommendations and 'informed choice' certainly presents a research opportunity to discover those who would most benefit from vaccination 26-45, but this grant seems to choose instead to focus on barriers to vaccination rather than those who would most benefit.

15. Key Strengths.

1. Important Clinical Problem (HPV+ OPC)
2. Team with necessary expertise in clinical domain and modeling

16. Key Weaknesses.

1. Conceptual gaps in how Aim 1<->3 effectively flow together
2. Clarity around association vs prediction modeling in Aim 2 needed
3. Aim 1 seems to focus more on barriers to vaccination rather than identification of most appropriate patients that might benefit from vaccination in ages 26-45, which is an active question.

CRITIQUE 3

1. Significance.

The problem statement is focused on the following:

1/Veterans have a higher and accelerating incidence rate of OPC and a higher rate of smoking than the general population.

2/ Smoking can make tx more complicated and decrease OS.

3/ together these trends portend an increased need for advocating for HPV vaccination to reduce the disease incidence.

4/ yet for those who develop HPV-OPC where tx is focused on de-intensification of therapy and minimally invasive surgical approaches to limit the morbidity of treatment and improve functional outcomes.

5/ Whereas those OPC patients with a smoking history de-escalation may be unsafe.

6/ They contend that while life-saving methods of primary (HPV vaccination) and secondary (smoking cessation) prevention methods are available, there are currently no tools in place to understand the current and future burden of OPC among Veterans to guide intervention target and set up policy prioritization.

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The Investigators published an analysis of SEER and VA MedSAS data which indicated that the increase in OPC incidence in the Veteran population may be 3 times greater than that at the national level.

This work revealed a total of 12,125 patients with oropharyngeal cancer identified from 10.9 million patients receiving care within the VHA health care system between 2000 and 2012 (a prevalence rate of 0.00111=111 per 100,000). Overall incidence of 12.2 cases of 100 000 persons, with a decrease in OPSCC incidence from 13.1 of 100 000 in 2000 to a low of 9.6 of 100 000 in 2006. This trend was followed by an increase in OPSCC incidence to a high of 15.6 of 100 000 in 2012.

Comparison to other conditions Veterans face is not made (Minor)

It is not clear what is meant by de-intensification or how it is measured (Moderate)

Not clear what is meant by tools beyond using the existing data available and sharing an algorithm for specific measures constructed. What type of tool is proposed? Will the team develop a dashboard or software or app? (Moderate)

A notable omission is any reference to the work by Clark et al and others about prevalence of OPC in more recent years among Veterans and the improved survival due to use of aspirin and anticoagulants. Clark et al (2021) reported on over 23000 Veterans with OPC from 2005 to 2018 Clark CM, Newark AC, Fokar A, Maxwell JH. Aspirin use predicts prolonged survival in patients with oropharyngeal cancer: Nationwide Veterans Affairs database study. Head Neck. 2021 Jan;43(1):247-254. doi: 10.1002/hed.26481. Epub 2020 Sep 22. PMID: 32959950. (Minor)

For reference the NCI reported overall cancer incidence rates (per 100 000 population) for all ages during 2013-2017 were 487.4 among males and 422.4 among females.

As a retrospective study using existing VA and NCDB data from 2018-2023, this study proposes that it will fill a gap about the future impact of HPV and HPV related OPC on the Veteran population.

Using existing VA data to document the prevalence of OPC and whether it is HPV related may be possible. However, it is not clear the team has the requisite experience and knowledge about most current VA information systems and data resources to construct a clinically meaningful cohort to address the questions. For example, whether a specific VHA CDW domain or fact table or other proposed data for use in this project include HPV vaccination data, a central measurement for the study, is never addressed. Nor is it explained if there is specific and sensitive test for distinguishing HPV and non-HPV related OPC and if it resides in the VHA data. (Major)

Significance section highlights preliminary research the team has published that shows that AA Veterans have equivalent treatment in VHA for OPC. Yet they propose that if vaccination for HPV is not better targeted that there could be disparate impacts of OPC. This point is not later addressed (Minor)

2. Innovation and Impact.

Innovation section describes that the study will focus on examining barriers to HPV vaccination. Yet the only data sources that will inform this question are from the VA CDW; there are no patient generated data included in the study design. (Moderate)

The proposal includes statistics on the Veteran population (Page 19) regarding the rate of HPV oncogenic infections and acknowledges that active-duty military are not required to have HPV

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vaccination. It is never described if VA data includes HPV vaccination data in a reliable form. The proposal needs to address this issue. (Major)

The proposal refers to a single site study focused on the Bronx VA study for methods (i.e., Ref 25) which did not use CDW; the study used only CPRS data and noted several limitations of this approach. It needs to be clarified if the proposed study intends to use any CPRS data like the Bronx study, in addition to the CDW and other data described in Methods. (Moderate)

Part of the study builds on a data set from 8 VHA facilities, yet the information contained in these data are not described to determine the feasibility of the data to address a robust set of measures to address the main study premise of barriers to HPV vaccination. These data appear to be unique and heretofore not described. A table of data sources and variable constructs should be included in the methods section. (Major)

The analytic methods proposed (linear regression, Markov modeling) are not innovative and no specific tool development is described. (Minor)

The patient experience section speaks about the potential ways they desire to engage Veterans for each Aim. Yet there are no firm commitments to carrying out the engagement activities mentioned. The proposal suggests the team will work with the VISNs to develop focus groups and "seek input from Veterans regarding attitudes on treatment deintensification". This goal seems out of line with the rest of the proposal, since treatment deintensification is never mentioned in later sections of the proposal. Further there is no mention of these activities, nor any credentials of investigators described to perform these qualitative analysis activities in the project plan. The patient engagement goals are aspirational at best. The proposal should be clear if patient engagement will be carried out in the project timeline and plans or if the information and knowledge gained in this study will inform a future project. (Moderate)

3. Approach.

The overall goal of the proposed study is described to understand the current and future burden of HPV (positive) related OPC in the VHA population by examining barriers to HPV vaccination, measuring disease incidence and outcomes in a national VHA HPV related OPC cohort, and developing a tool to model the potential benefit of primary and secondary prevention on HPV related OPC incidence and mortality. However, the Aims do not quite match this goal. Aim1 focuses on studying the prevalence (not the incidence) of HPV from 2018-2023 and identifying those populations that are under-vaccinated. Barriers studied are limited to that included in the VHA data and no patient generated information on barriers or facilitators are included. Aim 2 focuses on examining the role of smoking in contributing to outcomes of HPV-related OPC in a relatively small cohort from a database comprising patients from eight VAMCs. And Aim 3 is focused on predicting future HPV-related OPC incidence and cancer-related mortality in the VHA. The overall goal and Aims should be more clearly aligned. (Moderate)

For Aim 1, the study is appropriate as a retrospective study. No comparison group is proposed.

Regarding the adequacy of the data for the proposed study, the methods indicate a plan to use the CDW to identify HPV vaccination, but no information or preliminary analyses are described or shown to indicate if HPV vaccination data are reliably included in the VA CDW. The proposal lacks detail about the completeness of the data to construct the measures. For example, the proposal does not describe the data domains from which the vaccination data are present nor any preliminary data about their completeness. There is no mention of any limitations of the data such as if patients receive vaccinations from outside VA (e.g., community pharmacies or VA purchased care data or under other

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health insurance programs (e.g., Military Health system, Medicaid, Medicare), how such information is documented. There does not appear to be any reliance on text data from CPRS but may be necessary for historical information. The proposal should be clear about the data sources and the completeness of HPV vaccination data since this is a primary outcome measure. (Major)

The study cohort focuses on those engaged in VA care, i.e. those already known to VHA primary care; it excludes Veterans who have episodic connection to VHA. The study design should address if this limits the generalizability of the results. (Minor)

For the period of study 2018-2023, which includes the COVID-19 pandemic and implementation of CERNER, the study plan should address if these major events impacting Veterans healthcare use and data availability, respectively will affect the study. (Minor)

The sources of data were mentioned in the description of some variables, yet a mix of data sources were mentioned. For example, the proposal describes that to classify race and ethnicity they will rely on methods developed by VIREC reported in 2016 for using CDW. Yet they refer also to using the MEDSAS data instead of CDW. Other sections also mention use of MEDSAS for Aim 1/Experiment 1.2 and for Aim 2. The proposal should include a table of measures and data sources. (Moderate).

Aim 2 experimental approach is relying on a data set that appears to have already been created from 8 VHA institutions. If this exists it should be clearly described and included in a table of data sources (Moderate)

Aim 2 Statistical analyses refers to experiments 1.1 and 1.2; This section was confusing and may instead refer to experiment 2.1. The proposal should correct or clarify what is intended. (Minor)

Aim 2 methods experiment 2.1 focuses on smoking and OPC . The plan describes splitting the sample in thirds to allow for a validation component. The dataset appears to have only 300 HPV OPC patients between 2010-2018. The proposal should describe the generalizability of this sample. (Moderate)

The team has performed analyses of HPV OPC 2000-2015. The authors acknowledge some limitations of the patient reported smoking variable and suggest that they may also review chest imaging data as an alternative to patient reported pack years. The plan should describe further what would be reviewed and how it impacts the study timeline. (Minor)

Aim 3 methods, seems to link HPV OPC Incidence (note prevalence) with cancer related mortality. It's unclear if cause of death necessary for distinguishing cancer related death from general mortality will be included and added to the table of data sources (Moderate). Also the likely decreasing sample size for this Aim may have implications for generalizability and should here needs to be addressed (Moderate)

The simulation modeling planned will include a cohort of Veterans who utilized VA healthcare in 2008-2009 without a diagnosis of HPV (+) OPC) with demographics and characteristics drawn from the distributions of those of the Veteran population. It is described that the transition probabilities will vary by time/Markov cycle/age, because all of them are a function of age and other factors. The HPV-OPC probabilities seem to be drawn from information gleaned from Aim 1 or Aim 2, although the data are from a different era (2018-2023) than that proposed in Aim 3. The source of these transition probabilities for HPV and OPC should be clarified and explained (Moderate)

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Aim 3 Markov model for Veterans will also rely on published data for QALYs from the CEA registry. The authors need to describe these data wrt to Veterans specifically in this time era and population. If these data are not specific to Veterans, what are the implications for the estimates generated using these data? (Major)

4. Feasibility.

The sample and inclusion/exclusion for each Aim is described. Aim 1 will be limited to Veterans engaged in VHA primary care. For prevalent patients with OPC, since they may also be receiving care outside the VA for their specialty care, the plan should address the extent to which such patients may be excluded due to the study design. For Aims 2 and 3, generalizability of the study population to the overall Veteran population should be more clearly described. (Moderate).

Regarding the project timeline, it is proposed to be four years. Given that the data for Aims 2 and 3 already exist, this timeline seems excessive, and the timeline is unclear regarding whether each Aim is dependent on each other. For example, if Aim 2 relies on data and information from Aim 1 as described, it should begin near the end of Aim 1 work rather than completely overlapping as in the current plan. The proposal needs to revise the timeline and shorten it where possible. (Moderate)

5. Implementation.

There are no specific implementation or operations plans in the project timeline.

6. Investigator Qualifications.

The Investigators include Dr. Zevallos (PI) and Dr. Sandulache (CoI) who are clinical scientists in head and neck cancer epidemiology and clinical outcomes. Dr. Mazul (CoI) is an epidemiologist with expertise in head and neck cancer disparities. Dr. Chang (CoI) is a health economist and econometrician with experience using VA data and other data sources to conduct simulation studies to estimate life years lost and lifetime healthcare costs associated with obesity and obesity-related diseases and malignancies, using decision tree, Markov, Discrete Event Simulation, System Dynamics models. She will lead Aim 3 markov modeling.

Collectively, the team has collaborated on several projects on HPV related oropharyngeal cancer specific to the Veteran population including experience working with VHA administrative and clinical data through 2012. The team also has experience working on non-VA data sources such as SEER and the National Cancer Database. (Major)

The research timeline proposed is 4 years. Given that the data for Aims 2 and 3 already exist, this timeline seems excessive. The proposal should reconsider this timeline. (Moderate)

The team has experience and has published on a range of topics using the VHA data up through 2012; but it is not clear if there is experience using the CDW relational database and other data sources on VINCI. The project roles propose 4 full time staff as yet un-named to perform all the hands-on work with the data. It is unclear if any of the investigators have the requisite experience to train and oversee the extensive data management to prepare the data for analysis in the specified timeline. The proposal should clarify their experience working directly with VINCI and VHA CDW data for constructing similar data in more recent years and name specific staff with the requisite experience to carry out the work and address these issues. (Moderate)

ZEVALLOS, J

7. Multiple PI Leadership Plan.

Not applicable.

8. Facilities and Resources.

F& R appear adequate

Its noted that all data management and analyses will be performed on VINCI or on local VA computer systems.

9. Adequacy of Response to Previous Feedback Regarding the Proposed Study.

Not applicable.

10. Protection of Human Subjects from Research Risk.

Not applicable.

11. Inclusion of Women and Minorities in Research.

Not applicable.

12. Budget and Period of Support.

As noted previously the budget and timeline should be re-evaluated for fewer FTE and shorter timeline.

13. Sharing Research Data (Data Management and Access Plan (DMAP)).

Adequate.

14. Overall Impression.

This is a knowledgeable and experienced team; yet the quality of the proposal is lacking in detail to assess the value of the work planned.

Overall, it is unclear if the team has experience with the more current VA data structures and if any data from DOD or VA electronic health records in CPRS will be needed to construct some of the measures mentioned. Lack of details about the data sources to support construction of the main variables of interest and analyses makes it difficult to evaluate feasibility of the plans. The project will rely on existing albeit complex VA and other data; however, it should not warrant 4 years with 4 full time staff in addition to the PI/Col to perform the work. There is concern that with 4 TBN staff their training needs will delay progress. No publication plan is described

15. Key Strengths.

1. Experienced and knowledgeable investigators who have worked on similar projects together
2. Important research question to understand HPV vaccination and incidence and prevalence of OPC and HPV related OPC

ZEVALLOS, J

3. The investigators have published on these topics and generated some of the statistics on this using VA data through 2012.
4. Proposes to use existing data to address interesting research questions

16. Key Weaknesses.

1. Details about the data sources and feasibility of constructing the measures of interest is lacking
2. The relationship of the Aims to each other is not well described.
3. The staffing and timeline seem high/long for the work proposed.
4. Unclear if the team is sufficiently experienced in working with more recent CDW and other data to carry out the plan.

Footnotes for 1 I01 HX003625-01; PI Name: Zevallos, Jose P

+ Derived from the range of percentile values calculated for the study section that reviewed this application.

MEETING ROSTER

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Health Services Research Parent IRG
Office of Research & Development
HSR3
03/02/2022 - 03/03/2022**

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Consultants are required to absent themselves from the room during the review of any application if their presence would constitute or appear to constitute a conflict of interest.